

Math 571 - Homework 1

Problem 1.1 (R:1:2*). Show that for any positive integer n , if n is not a perfect square, then $\sqrt{n}$ is irrational.

Suppose $\sqrt{n} = p/q$ where p and q are integers with no common factors, i.e., $\gcd(p, q) = 1$. Then $n = p^2/q^2$ so $nq^2 = p^2$. But we know that $\gcd(p^2, q^2) = 1$ and that if $\gcd(a, b) = 1$ and $a|bc$, then $a|c$, thus $p^2|n$. This means $p^2 = nq^2 = n'p^2q^2$ and so $n'q^2 = 1$, thus $n' = 1$ and $q = 1$ hence $n = p^2$.

Problem 1.2 (R:1:4*). Let E be a non-empty subset of an ordered set $(S, <)$; suppose that α is a lower bound for E in S and β is an upper bound for E in S . Show that $\alpha \leq \beta$. Can $\alpha = \beta$? What happens if $E = \emptyset$?

As E is non-empty, let $s \in E$, then $\alpha \leq s \leq \beta$. It could be that $E = \{s\}$ and so $\alpha = s = \beta$. If $E = \emptyset$, then for $s \in S$, s is both a lower-bound and an upper-bound for E , thus if $|S| > 1$ it is possible that $\beta < \alpha$.

Problem 1.3 (R:1:5). Let A be a non-empty set of real numbers bounded below. Let $-A = \{-a \mid a \in A\}$. Show that

$$\inf(A) = -\sup(-A)$$

Let $\alpha = \inf(A)$. We have $\alpha \leq a$ for all $a \in A$ and thus $-\alpha \geq -a$ for all $a \in A$. So $-A$ is bounded above by $-\alpha$.

Suppose that β is any upper-bound for $-A$, then, as above, $-\beta$ is a lower-bound for A and hence $-\beta \leq \alpha$, but then $-\alpha \leq \beta$. Thus $-\alpha = \sup(-A)$. This yields the desired result.

The next two problems, R:1:6 and R:1:7, show that the exponentiation map from $\mathbb{Q}$ to $\mathbb{R}$ given by $r \mapsto b^r$ for $b > 1$ can be uniquely extended to $x \mapsto b^x$ for $x \in \mathbb{R}$ and that for $b > 1$ and $y \in (0, \infty)$ we can find x so that $b^x = y$, that is, $\log_b : (0, \infty) \rightarrow \mathbb{R}$ is well-defined.

Problem 1.4 (R:1:6). Fix $b > 1$.

(a) If n, m, p, q are integers, $n, q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Because of this defining b^r and $b^{m/n}$ for any $m/n = r$ provided $n > 0$ is well-defined.

Let $v = (b^m)^{1/n}$ and $w = (b^p)^{1/q}$. By the definition of fractional exponents, $v^n = b^m$ and $w^q = b^p$. Using the properties of integer exponents, we can raise both sides to find a

common exponent:

$$\begin{aligned} v^{nq} &= (v^n)^q = (b^m)^q = b^{mq} \\ w^{nq} &= (w^q)^n = (b^p)^n = b^{pn} \end{aligned}$$

We are given that $m/n = p/q$, which implies $mq = pn$. Therefore, substituting this into the exponent gives $b^{mq} = b^{pn}$, which means $v^{nq} = w^{nq}$.

Because $b > 0$, both v and w are positive. Since the map $x \mapsto x^{nq}$ is injective for positive reals (Theorem 1.21 on the uniqueness of positive roots), $v^{nq} = w^{nq}$ implies $v = w$.

Because of this, defining $b^r = b^{m/n}$ for any rational $r = m/n$ (with $n > 0$) is well-defined.

- (b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

Let $r = m/n$ and $s = p/q$, then

$$\begin{aligned} b^{r+s} &= b^{(qm+np)/qn} && \text{since } r+s = (qm+np)/qn \\ &= (b^{qm+np})^{1/nq} && \text{by (a)} \\ &= (b^{qm} b^{np})^{1/nq} \\ &= (b^{qm})^{1/nq} (b^{np})^{1/nq} && \text{since } (ac)^{1/k} = a^{1/k} c^{1/k} \\ &= b^r b^s && \text{by (a)} \end{aligned}$$

Here we do use that $a^{1/k} c^{1/k} = (ac)^{1/k}$ this is easily shown by

$$\begin{aligned} a^{1/k} = \alpha \text{ and } c^{1/k} = \gamma &\iff a = \alpha^k \text{ and } c = \gamma^k \\ &\iff ac = \alpha^k \gamma^k = (\alpha\gamma)^k \\ &\iff (ac)^{1/k} = \alpha\gamma = a^{1/k} c^{1/k} \end{aligned}$$

- (c) If $x \in \mathbb{R}$, define $B(x) = \{b^t \mid t \in \mathbb{Q} \wedge t \leq x\}$. Prove that

$$b^r = \sup(B(r))$$

whenever r is rational.

Explain why it makes sense to define

$$b^x = \sup(B(x))$$

for every real x .

Suppose $m/n = s < r = p/q$ are rational with $n, q > 0$, then $mq < np$ and

$$b^{mq} < b^{np} \implies b^{mq/nq} < b^{np/nq} \implies b^s < b^r$$

This uses the fact that $0 \leq a < c \implies a^{1/k} < c^{1/k}$ for k a positive integer, i.e., the k^{th} -root function is *increasing* on $[0, \infty)$.

So b^r is an upper bound of $B(r)$ and since $b^r \in B(r)$ we have that b^r is the least upper bound, in fact, b^r is the maximal member of $B(r)$, so $b^r = \sup(B(r))$.

We know $B(x)$ is bounded above for each $x \in \mathbb{R}$ since if $r \in \mathbb{Q}$ and $r > x$ we have $b^r \geq B(x)$. So $b^x = \sup(B(x))$ exists. We have also seen that defining b^x in this manner extends the map $r \mapsto b^r$ for $r \in \mathbb{Q}$. Later, we will see that this is the *unique* continuous extension to $\mathbb{R}$ of this map.

- (d) Prove that $b^{x+y} = b^x b^y$ for every real x and y .

First, we show $\sup B(x+y) \leq \sup B(x) \cdot \sup B(y)$. Let a be any real number such that $a < \sup B(x+y)$. By the definition of supremum, a cannot be an upper bound for $B(x+y)$, so there must exist some rational $s \leq x+y$ such that $a < b^s \leq \sup B(x+y)$. Because $b > 1$ and the rationals are dense, we can find a rational $t < s$ such that $a < b^t < b^s$. Since $t < s \leq x+y$, we have the strict inequality $t < x+y$.

Key Step: We want to make the seemingly obvious move that as $t < x+y$ we can write $t = t_x + t_y$ for some $t_x < x$ and $t_y < y$ where $t, t_x, t_y \in \mathbb{Q}$. Once we have this, we have $b^t = b^{t_x} b^{t_y}$. Since $t_x < x$, $b^{t_x} \in B(x)$, so $b^{t_x} \leq \sup B(x)$. Since $t_y < y$, $b^{t_y} \in B(y)$, so $b^{t_y} \leq \sup B(y)$. Therefore, $a < b^t \leq \sup B(x) \cdot \sup B(y)$. Because every real number $a < \sup B(x+y)$ is strictly less than $\sup B(x) \sup B(y)$, it must be that $\sup B(x+y) \leq \sup B(x) \cdot \sup B(y)$.

Claim: If $t \in \mathbb{Q}$ and $t < x+y$, there is $t_x, t_y \in \mathbb{Q}$ so that $t_x < x$, $t_y < y$, and $t = t_x + t_y$.

Proof of the claim: Let $\epsilon = (x+y) - t > 0$. By the density of $\mathbb{Q}$, we can choose a rational t_x such that $x - \epsilon < t_x < x$. Let $t_y = t - t_x$. This is rational, and $t_y = t - t_x < t - (x - \epsilon) = t - x + (x + y - t) = y$. Thus, $t = t_x + t_y$ where $t_x < x$ and $t_y < y$.

Next, we show the reverse inequality: $\sup B(x) \cdot \sup B(y) \leq \sup B(x+y)$. Take $a < \sup B(x) \cdot \sup B(y)$. Then $\frac{a}{\sup B(x)} < \sup B(y)$. By the definition of supremum, $a/\sup B(x)$ is not an upper bound for $B(y)$, so there exists some $t_y \in \mathbb{Q}$ with $t_y \leq y$ such that $\frac{a}{\sup B(x)} < b^{t_y}$. Rearranging gives $\frac{a}{b^{t_y}} < \sup B(x)$. Applying the definition of supremum again, there exists some $t_x \in \mathbb{Q}$ with $t_x \leq x$ such that $\frac{a}{b^{t_y}} < b^{t_x}$. This means $a < b^{t_x} b^{t_y} = b^{t_x+t_y}$. Since $t_x \leq x$ and $t_y \leq y$, we have $t_x + t_y \leq x+y$, which means $b^{t_x+t_y} \in B(x+y)$. Thus, $a < b^{t_x+t_y} \leq \sup B(x+y)$. Since this holds for any $a < \sup B(x) \cdot \sup B(y)$, we conclude $\sup B(x) \cdot \sup B(y) \leq \sup B(x+y)$.

Problem 1.5 (R:1:7). Fix $b > 1$, $y > 1$, and show that there is a unique x so that $b^x = y$. This defines $\log_b(y) = x$.

- (a) For any positive integer n , $b^n - 1 \geq n(b - 1)$.

This is easy to see as

$$\frac{b^n - 1}{b - 1} = \sum_{i=0}^{n-1} b^i \geq \sum_{i=0}^{n-1} 1 = n$$

- (b) $b - 1 \geq n(b^{1/n} - 1)$.

Let $c = b^{1/n}$, then by (a), since $c > 1$,

$$c^n - 1 \geq n(c - 1), \text{ so } b - 1 \geq n(b^{1/n} - 1)$$

- (c) If $t > 1$ and $n(t - 1) > (b - 1)$, then $b^{1/n} < t$.

From (a) we have $t^n - 1 \geq n(t - 1) > b - 1$, so $t^n > b$ and thus $t > b^{1/n}$ since b^x is monotonic increasing.

- (d) If w is such that $b^w < y$, then $b^{w+1/n} < y$ for large enough n . To see this, apply (c) to $t = y \cdot b^{-w}$.

Let $t = y \cdot b^{-w}$, note that $t > 1$ by assumption. By the Archimedean Property (Theorem 1.20), take n large enough so that $n(t - 1) > b - 1$. By (c), $b^{1/n} < t = y \cdot b^{-w}$ and thus, $y > b^w b^{1/n} = b^{w+1/n}$. This is what we wanted.

- (e) If $b^w > y$, then $b^{w-1/n} > y$ for large enough n .

Imitate the above, let $t = y^{-1} \cdot b^w > 1$ and let n be large enough so that $n(t - 1) > b - 1$, then again by (c), $b^{1/n} < t = y^{-1} \cdot b^w$. Thus we get $y^{-1} < b^{-w} b^{1/n}$ and so $y > b^w b^{-1/n} = b^{w-1/n}$.

- (f) Let $A(y) = \{w \mid b^w < y\}$ and $x = \sup(A(y))$. Show that $y = b^x$.

Before starting this argument, notice that $A(y)$ is the *obvious* set to try to use here. The argument below indicates how (d) and (e) are natural. Thus, (a)–(c) can be seen as technical steps used to prove (d) and (e).

Clearly, $A(y)$ is bounded above so that x exists. $y \leq b^x$ since otherwise, $b^x < y$ hence $b^x \in A(y)$. By (d) this can't happen as $b^{x+1/n} < y$ for large enough n , so $x + 1/n \in A(y)$ contradicting $x = \sup(A(y))$.

If $y < b^x$, then by (e) $b^{x-1/n} > y$ for large enough n . But $x - 1/n < \sup(A(y))$ and so $b^{x-1/n} < y$. Thus $b^x \leq y$.

We have shown $b^x \leq y \leq b^x$ and so $y = b^x$.

- (g) Show that x is unique.

Proof 1: This question really could, and should, be part of (1.6), as it follows from the fact that $x \mapsto b^x$ is one-to-one. In fact, let us see that $x \mapsto b^x$ is strictly increasing. Suppose $x < z$ and fix $s \in \mathbb{Q}$ with $x < s < z$. Let $t, r \in \mathbb{Q}$ be such that $t < x < s < r < z$. It is clear that $b^t < b^s < b^r$, since we know that exponentiation $r \mapsto b^r$ is increasing on $\mathbb{Q}$. Since t and r are arbitrary, $b^x = \sup B(x) \leq b^s < \sup B(z) = b^z$.

Proof 2: Suppose $b^x = b^z$, then from (R:1:6 (d)) $b^x b^{-z} = b^{x-z} = 1$. It suffices to show $w = x - z = 0$.

Suppose $w > 0$, then let $0 < r < w$ with $r \in \mathbb{Q}$. Take s with $r < s < w$ and $s \in \mathbb{Q}$, then $1 < b^r < b^s \leq \sup B(w) = b^w$.

Suppose $w < 0$, then let $w < r < 0$ with $r \in \mathbb{Q}$. Let $s \in \mathbb{Q}$ with $w < s < r < 0$. We have that for all $t \in \mathbb{Q}$, $t < w \implies b^t < b^s$ and so b^s is an upper bound of $B(w)$ and hence $b^w = \sup B(w) \leq b^s < b^r$. Since $b > 1$, $b^{-r} > 1$ so $b^r < 1$ and $b^w < 1$.

Problem 1.6 (R:1:8). Show that $\mathbb{C}$ can not be made into an ordered field.

We show that in an ordered field, $a^2 \geq 0$ for all a . This is by definition for $a > 0$ and trivial for $a = 0$, so we need to see that it holds for $a < 0$.

If $a < 0$, then $-a > 0$, this is because $a > 0 \implies 0 = a + (-a) > 0 + (-a) = -a$. So $(-a)(-a) > 0$, if we can just show $\boxed{(-a)(-a) = a^2}$, then we are done. This is part of Theorem 1.18, but let's give an argument anyway.

For this it would be nice to argue $\boxed{(-a) = -1(a)}$ so that $(-a)(-a) = (-1)^2 a^2$. Then we just need to see that $\boxed{(-1)^2 = 1}$. If we knew $\boxed{0a = 0}$, then $(1 + (-1))a = 0$ so $1a + (-1)a = a + (-1)a = 0$ and by the uniqueness of inverses, $(-1)a = -a$.

Since

$$0 = (-1)(0) = (-1)(1 + (-1)) = -1(1) + (-1)(-1) = -1 + (-1)^2,$$

we have, by uniqueness of inverses, $\boxed{(-1)^2 = 1}$.

Finally, since $0a + a = 0a + 1a = (0 + 1)a = 1a = a$ we have $0a + a = a$. Adding $-a$ to both sides gives $\boxed{0a = 0}$.

The following two problems are also paired. R:1:14 is really just a special instance of R:1:17*, which differs from Rudin by including the case $\mathbf{x}, \mathbf{y} \in \mathbb{C}^k$

Problem 1.7 (R:1:14). Show that for $w, z \in \mathbb{C}$

$$|w + z|^2 + |w - z|^2 = 2|w|^2 + 2|z|^2.$$

Use this to compute $|1 + z|^2 + |1 - z|^2$ given that $|z| = 1$.

Getting $|1 + z|^2 + |1 - z|^2 = 2|w| + 2|z| = 2 + 2 = 4$ given that $|z| = 1$ is trivial by letting $w = 1$.

For the main part we have

$$\begin{aligned} |w + z|^2 + |w - z|^2 &= (w + z)\overline{(w + z)} + (w - z)\overline{(w - z)} \\ &= (w + z)(\bar{w} + \bar{z}) + (w - z)(\bar{w} - \bar{z}) \\ &= w\bar{w} + z\bar{w} + w\bar{z} + z\bar{z} + w\bar{w} - z\bar{w} - w\bar{z} + z\bar{z} \\ &= 2|w|^2 + 2|z|^2 \end{aligned}$$

Problem 1.8 (R:1:17*). Show that for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k(\mathbb{C}^k)$,

$$\|\mathbf{x} + \mathbf{y}\|_2^2 + \|\mathbf{x} - \mathbf{y}\|_2^2 = 2\|\mathbf{x}\|_2^2 + 2\|\mathbf{y}\|_2^2. \quad (\text{Parallelogram Law})$$

How does this generalize the Pythagorean theorem?

This is proved exactly as in the previous problem. The operation corresponding to conjugate above is the *Hermitian* or *Conjugate Transpose* (See [here](#)). $A^H = (\bar{A})^T = \overline{A^T}$

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|_2^2 + \|\mathbf{x} - \mathbf{y}\|_2^2 &= (\mathbf{x} + \mathbf{y})^H (\mathbf{x} + \mathbf{y}) + (\mathbf{x} - \mathbf{y})^H (\mathbf{x} - \mathbf{y}) \\ &= (\mathbf{x}^H + \mathbf{y}^H)(\mathbf{x} + \mathbf{y}) + (\mathbf{x}^H - \mathbf{y}^H)(\mathbf{x} - \mathbf{y}) \\ &= \mathbf{x}^H \mathbf{x} + \mathbf{x}^H \mathbf{y} + \mathbf{y}^H \mathbf{x} + \mathbf{y}^H \mathbf{y} + \mathbf{x}^H \mathbf{x} - \mathbf{x}^H \mathbf{y} - \mathbf{y}^H \mathbf{x} + \mathbf{y}^H \mathbf{y} \\ &= 2\|\mathbf{x}\|_2^2 + 2\|\mathbf{y}\|_2^2 \end{aligned}$$